Thrinadh Sai Nimmagadda

8033548350 thrinadhnimmagadda20@gmail.com Linkedin/Thrinadh Sai Nimmagadda GitHub/

Machine Learning Engineer

Experienced machine learning engineer skilled in building scalable AI systems using Python, NLP, and LLMs. Proven success developing RAG pipelines, semantic search, and real-time anomaly detection with strong backend API and microservices expertise. Demonstrated ability to optimize model performance, reduce latency, and implement privacy-safe AI solutions. Adept at full-stack development, data engineering, and delivering actionable insights with advanced analytics.

SKILLS

Programming Languages: Python, Java, SQL, JavaScript, C++.
Machine Learning & NLP:LLMs, RAG Pipelines, NLP, XGBoost, scikit-learn, TensorFlow, PyTorch, SHAP, Sentence-Transformers.
Frameworks & Libraries: FastAPI, Django, Flask, Spring Boot, React, LangChain, LangGraph, Hugging Face, FAISS.
Databases & Tools: PostgreSQL, MySQL, Redis, Docker, Git, CI/CD, Kafka, ChromaDB, Power BI.

PROFESSIONAL EXPERIENCE

AI Intern
Artificial Intelligence Institute of South Carolina May 2025 - Aug 2025 | USA

- Engineered an NLP clustering pipeline leveraging Sentence-BERT and UMAP to organize over 100K research papers, enhancing semantic search capabilities beyond keyword matching.
- Integrated a RAG chat feature using chunked PDF embeddings and a local LLaMA model, reducing per-query token costs while enabling detailed paper-specific Q&A.
- Constructed REST APIs with Django REST Framework and developed an interactive React frontend, improving data accessibility and user engagement with clustered document insights.
- Accelerated query response times from minutes to seconds by implementing asynchronous processing and caching for scalable semantic search clusters.

Software Engineer
Wipro Jul 2021 - Aug 2024 | India

- Enhanced resume parsing accuracy from 50% to 80% by implementing secondary extraction passes identifying skills within job descriptions and experience text.
- Optimized API latency by 20% through refining response workflows and batching database writes, improving overall system performance and throughput.
- Rectified extraction issues in multi-column PDF resumes by grouping text blocks horizontally and reconstructing reading order for reliable data extraction.
- Standardized formatting and resolved broken line breaks across PDF and Word resumes, boosting document processing consistency and backend integration reliability.

PROJECTS

Incident Copilot

- Engineered an AI agent workflow leveraging MCP tools for anomaly investigation and log search to enable real-time incident detection.
- Instituted deterministic distance-threshold guardrails to eliminate false-positive matches during adversarial testing, enhancing system reliability.
- Validated the system with 21 automated tests ensuring robustness in Kafka parsing, anomaly detection, and AI agent workflows.

ClinicalMatch AI

- Constructed a clinical trial matching system integrating BM25 keyword search with semantic retrieval to elevate recall on non-small cell lung cancer trials.
- Devised a corrective RAG loop that autonomously refines queries with expanded clinical terminology to improve retrieval accuracy on weak initial results.
- Optimized operational costs by 13.2x through eligibility prefilters and caching strategies while retaining top-tier LLM reasoning quality.

CogniTeam

- Formulated ML pipelines for predicting burnout, attrition, and conflict risk using XGBoost and NetworkX-based communication signals.

- Instituted GenAI guardrails to prevent sensitive data exposure and restrict punitive or medical/legal overreach in employee intelligence insights.
- Delivered a privacy-safe RAG assistant combining SQL-grounded risk metrics with authoritative HR and regulatory guidance using a locally hosted LLaMA model.

EDUCATION

MS in Computer Science(CGPA: 3.5)
University of South Carolina — Aug 2024 - Dec 2026 | South Carolina,USA

Bachelor's in Computer Science(CGPA: 4.0)
B.V. Raju Institute of Technology — Jul 2018 - Jun 2022 | India

Certification
Azure Fundamentals
Credential ID AB31F3DFF1D78F8C